United International University

Department of Computer Science & Engineering

Final Examination, Trimester: Summer 2025

Course Code: CSE 3715 **Course Title:** Data Communication

Duration: 2 hours **Total Marks:** 40

(Any examinee found adopting unfair means would be expelled from the trimester/program as per UIU disciplinary rules.)

ANSWER ALL THE QUESTIONS

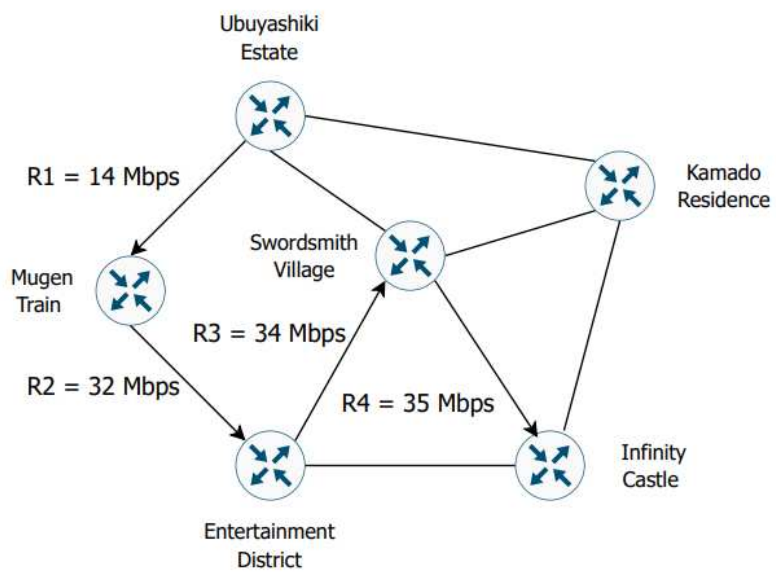
1. A software company needs to connect its two offices, one in Ubuyashiki Estate and another in Infinity Castle — for real-time video conferencing and large file transfers. The network team is evaluating whether to use a circuit-switched or packet-switched communication model.

   (a) **Compare** the both approaches in terms of delay, resource utilization and quality of service. **[3] (CO1)**

   (b) **Suggest** the most suitable switching method for each service and **justify** your answer. **[2] (CO1)**

   (c) Consider the network shown below, where Ubuyashiki Estate is the source and Infinity Castle is the destination. Data is transmitted using the packet switching method, following the arrowed path. Each router introduces a processing and queuing delay of 5 ms. The transmission rates of the links are denoted by R1, R2, R3, and R4. **[5] (CO1)**

   If 201 bytes of data are sent, **calculate** the end-to-end delay. Assume the distance between each node is 500 km and the propagation speed of the medium is $2 \times 10^8 ms^{-1}$.



2. Ubuyashiki children, the same network team from Question 01, are now trying to test different methods to ensure reliable data transmission over their network. During the tests, they observed that noise in the communication channel occasionally altered a few bits of the transmitted data. To evaluate the reliability of their network, they are applying various error detection and correction techniques.

(a) The following bit stream is to be sent using checksum error detection with 8 bits per block: **[2] (CO5)**

11001100 10101010

Calculate the encoded data by adding the checksum to the message for transmission.

(b) The same system is then tested using CRC error detection with the generator polynomial, $G(x) = x^5 + x^4 + x^2 + 1$. If the received data is: **[4] (CO5)**

$$x^{14} + x^{12} + x^8 + x^7 + x^5 + x^3 + x^2 + x$$

Perform the error detection at the receiver end and **conclude** whether the received frame will be accepted or rejected.

(c) The engineers now encode small data blocks into 7-bit sequences using Hamming Code C(7,4). One received 7-bit codeword is: **[4] (CO5)**

1011011

Detect and **correct** any error, if present, and **write** the original 4-bit data.

3. In a wireless LAN, multiple devices share the same channel using a random multiple access technique to avoid collisions. To minimize the chances of two stations transmitting at the same time, this method makes each station sense the channel before sending data and use an acknowledgment-based feedback system. However, unlike wired networks, this system cannot detect collisions while transmitting, which often leads to challenges such as the hidden station problem.

   (a) **Identify** and elaborately **explain** the random multiple access technique used in this wireless network. **[5] (CO4)**

   (b) **Explain** why collision detection is not practical in wireless communication, even though it is used in wired networks. **[2] (CO4)**

   (c) **Describe** the hidden station problem and explain how it can be solved. **[3] (CO4)**

4. Citycell was a well-known operator offering clear voice quality and wide coverage, during the 2G mobile communication era in Bangladesh. Unlike other companies, its service did not use removable SIM cards — users needed special phones that worked only on its network. Over time, as mobile technology and user preferences evolved, Citycell's system failed to keep pace and eventually disappeared from the market.

   (a) **Describe** the network architecture used by the SIM-based operators of that period. **[4] (CO4)**

   (b) **Compare** Citycell's communication method with the SIM-based systems in terms of channel access technique, user identity, and scalability. **[3] (CO4)**

   (c) **Explain** why Citycell's technology declined despite its technical advantages. **[3] (CO4)**